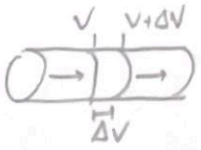
PFR & PBR

$$q - F_{A0}\frac{dT}{dV}\left(\bar{C}_{P_0} + \Delta\tilde{C}_p X_A\right) - F_{A0}\frac{dX_A}{dV}\left(\Delta H^\circ_{rxn} + \Delta\hat{C}_p(T-T_r)\right) = 0 \quad ; \quad F_{A0}\frac{dX}{dV} = -r_A(+,\bar{})$$



$$q = U_{av}(T_a - T) = U\left(\frac{4}{D}\right)(T_a - T)$$

Q_q = $r_A\Delta H_R$; Q_R = $U_a(T-T_a)$

$$(W_s = 0) \therefore \quad \frac{dT}{dV} = \frac{r_A\Delta H_R - U_a(T-T_a)}{\Sigma F_i C_{P_i}}$$

product rule

$$\begin{cases} Q\Delta V + \Sigma F_i H_i|_V - \Sigma F_i H_i|_{V+\Delta V} = 0 \\ U_a(T_a - T) - \dfrac{d\Sigma(F_i H_i)}{dV} = 0 \end{cases}$$

$$\frac{dT}{dV} = \frac{r_A\Delta H_{Rxn} - U_a(T-T_a)}{F_{A0}(\Sigma\Theta_i C_{P_i} + \Delta C_p X)} \quad \text{as } f(X)$$

For heat exchanger:

$$-F\frac{dH}{dV} = U_a(T_a - T) \; ; \; \frac{dH}{dV} = C_p\frac{dT}{dV}$$

$$\Rightarrow \frac{dT}{dV} = \frac{U_a(T_a-T)}{(FC_p)} \; ; \; F = \frac{\dot{m}}{MW}$$

PBR: $$\frac{dT}{dW} = \frac{r_A'\Delta H_R - \dfrac{U_a(T-T_a)}{\rho_b}}{\Sigma F_i C_{P_i}} \; ; \quad W = \rho_B V$$

Co-current $\Rightarrow \dfrac{dT_a}{dV} = \dfrac{U_a(T-T_a)}{\dot{m}_c C_{P_c}}$

Counter current $\Rightarrow \dfrac{dT_a}{dV} = \dfrac{U_a(T_a-T)}{\dot{m}_c C_{P_c}}$

Common (for all)

$$X^{EB} = \frac{\Sigma\Theta_i C_{P_i}(T-T_0)}{-\left[\Delta H^\circ_{A}(T_R) + \Delta C_p(T-T_R)\right]}$$

$$T = \frac{X\left[-\Delta H^\circ_{Rx}(T_R)\right] + \Sigma\Theta_i C_{P_i}T_0 + X\Delta C_p T_R}{\Sigma\Theta_i C_{P_i} + X\Delta C_p}$$

$\Delta C_p = 0$ ↓

$$\approx T_0 + \frac{X\left[-\Delta H^\circ_{Rx}(T_R)\right]}{\Sigma\Theta_i C_{P_i}}$$

(when $\Delta C_p = 0$ & feed is pure A)

Adiabatic

- $k = A_0 \exp\left(\frac{-E_a}{RT}\right)$
- $k = k_1(T_R)\exp\left(\frac{E_a}{R}\left(\frac{1}{T_R} - \frac{1}{T}\right)\right)$
- $K_c = K_c(T_1)\exp\left[\frac{\Delta H^\circ_R}{R}\left(\frac{1}{T_2} - \frac{1}{T}\right)\right]$
- $\Delta H_{Rxn} = \Delta H^\circ_{r_{am}}(T_R) + \Delta\tilde{C}_p(T - T_R)$
- $S_D = \frac{r_D}{r_u}$; $S_p = \frac{F_D}{F_u}$ (overall)

$$\frac{\partial E}{\partial t} = \dot{Q} + \dot{W}_s - F_{A0}\Sigma\Theta_i\tilde{C}_{P_i}(T-T_0) - F_{A0}X_A\left[\Delta H^\circ_{rxn} + \Delta\hat{C}_p(T-T_R)\right]$$

↳ This does not include preheating the feed as it is only concerned with T after all reactants have reached T_0

$\tilde{Y} = \frac{F_D}{F_{A0} - F_A}$ (Overall yield) ; $Y_{D/A} = \frac{r_D}{-r_A}$ (Inst yield)

Isothermal:

Δ Adiabatic

$Q_g = Q_r$

$Q_r \rightarrow UA(T-T_a)$

$Q_g \rightarrow \Delta H_{rxn}(r_A V)$

For a CSTR: $F_{A0}X\Delta H_{rxn}$